

NAME: _____

PERIOD: _____

DATE: _____

A Controller Comparison

AP Statistics · Unit 4 · Topic 4.8 · B: 2027-01-26 · E: 2027-03-15

[STAR] TODAY'S PROBLEM

Suppose a cooperative has 1,200 cases with conventional controllers and 900 with adaptive controllers. It takes independent SRSs of 40 existing cases from each group; controller type was not assigned. A 95% interval for $\mu_C - \mu_A$ is (1.219, 4.781) kWh per case-day. A retrofit is worthwhile only if mean savings are at least 2.0 kWh.

Rhea: “Zero is excluded and 2 is inside, so retrofitting is shown to cause savings of at least 2 kWh.”

Sol: “I would compare the interval separately with 0 and 2, then inspect sampling and assignment before applying it to a retrofit.”

Daily kWh samples

Controller	N	n	$\bar{x}$	s
Conv.	1,200	40	18.6	4.0
Adaptive	900	40	15.6	4.0

FIRST TAKE — before the video (5 min)

Which account is better supported so far? Identify any interval feature or design detail needed to decide.

[BOOK] CHECK ZERO, THE CLAIM'S THRESHOLD, AND THE DESIGN (keep on the page)

A confidence interval for $\mu_1 - \mu_2$ describes plausible values for a difference in population means in the stated subtraction order, not a range containing a percent of individual observations. If 0 is excluded, the interval supports a population mean difference; a practical claim such as $\mu_1 - \mu_2 \geq c$ requires the entire interval to lie at or beyond c . For a fixed interval, say “we are confident,” not “there is a probability.” Random sampling supports population scope, while random assignment is needed to isolate a causal treatment effect.

Finish after the video:

DOK 1 (a) Define $\mu_C - \mu_A$, calculate the point estimate, and locate 0 and 2 relative to the interval.

DOK 2 (b) Verify independent sampling, both 10 percent checks, and large-sample shape. Reproduce SE , df , margin of error, and the interval; interpret it.

DOK 3 (c) Adjudicate the accounts using the positions of 0 and the 2-kWh threshold. Bound population and causal scope from sampling and assignment, and explain the budgeting consequence of the rejected interpretation.

Frame: Relative to the interval, 0 is _____ and 2 is _____; those positions support _____.

Frame: The sampling design supports _____ but leaves _____ unresolved because _____.

EXIT — turn this in

One thing the video changed about my first take: _____